

Bank Customer Churn Prediction System

A Deep Learning Approach with PyTorch, SMOTE Class Balancing and an Interactive Streamlit Deployment

MODEL TYPE	Fully-Connected Neural Network (PyTorch)
DATASET	10,000 Bank Customer Records
TASK	Binary Classification (Churn / Stay)
REPORT DATE	September 26, 2026

Table of Contents

01 Executive Summary
02 Business Problem & Objective
03 Dataset Overview
04 Exploratory Data Analysis
05 Methodology & Preprocessing
06 Model Architecture
07 Training Configuration
08 Results & Evaluation
09 Application Deployment
10 Limitations
11 Conclusion & Future Work

01 Executive Summary

This report documents the design, training, and deployment of a machine learning system that predicts customer churn for a retail bank. Customer churn — when a customer closes their account or stops using a bank's services — is a critical business metric, since retaining an existing customer is significantly cheaper than acquiring a new one. The project delivers an end-to-end solution: a PyTorch neural network trained on 10,000 historical customer records, and a production-style Streamlit web application that allows non-technical users to query the model in real time or in bulk.



Key Highlights

- End-to-end pipeline: data preprocessing → class balancing → model training → evaluation → deployment.
- A 5-layer fully-connected neural network built and trained from scratch in PyTorch.
- Class imbalance (~20% churn rate) addressed using SMOTE oversampling.
- Recall of 96.5% on the test set — the model correctly flags the vast majority of customers who actually churn, which is the priority metric for a retention use case.
- Deployed as an interactive Streamlit dashboard with single-customer prediction, batch CSV scoring, and data-insight visualizations.

02 Business Problem & Objective

Customer attrition directly erodes a bank's revenue base and long-term customer lifetime value. Industry research consistently shows that acquiring a new customer costs substantially more than retaining an existing one, which makes early identification of at-risk customers a high-leverage business capability. Retention teams need a reliable way to identify, ahead of time, which customers are likely to leave so that targeted interventions — loyalty offers, personalized outreach, or service improvements — can be applied before the customer actually exits.

The objective of this project is to build a supervised binary classification model that takes a customer's profile (demographic and account information) as input and outputs a churn probability. This probability can then be used by the bank to rank customers by risk and prioritize retention efforts accordingly.

Project Goals

- Build an accurate churn classifier using a custom neural network rather than an off-the-shelf model.
- Handle the natural class imbalance present in churn data without sacrificing recall on the minority (churned) class.
- Package the model into a usable, interactive tool rather than leaving it as a notebook artifact.
- Document the full pipeline for reproducibility and portfolio purposes.

03 Dataset Overview

The model is trained on the **Bank Customer Churn Records** dataset, containing 10,000 rows, each representing a unique bank customer described by demographic, financial, and engagement attributes. The target variable, **Exited**, is binary: 1 if the customer churned, 0 if they remained a customer.

Feature Dictionary

Feature	Type	Description
CreditScore	Numerical	Customer's credit score
Geography	Categorical	Country of residence (France, Germany, Spain)
Gender	Categorical	Male / Female
Age	Numerical	Customer age in years
Tenure	Numerical	Years as a bank customer
Balance	Numerical	Current account balance
NumOfProducts	Numerical	Number of bank products used (1–4)
HasCrCard	Binary	Whether the customer holds a credit card
IsActiveMember	Binary	Whether the customer is an active member
EstimatedSalary	Numerical	Estimated annual salary
Exited (Target)	Binary	1 = churned, 0 = retained

Dataset Summary

Metric	Value
Total records	10,000
Churned customers	2,038 (20.4%)
Retained customers	7,962 (79.6%)
Numerical features	8
Categorical features	2 (one-hot encoded)
Missing values	None

04 Exploratory Data Analysis

Before training, the dataset was explored to understand churn drivers and validate feature relevance. The visualizations below summarize the most informative patterns found in the data.

Churn Rate by Geography

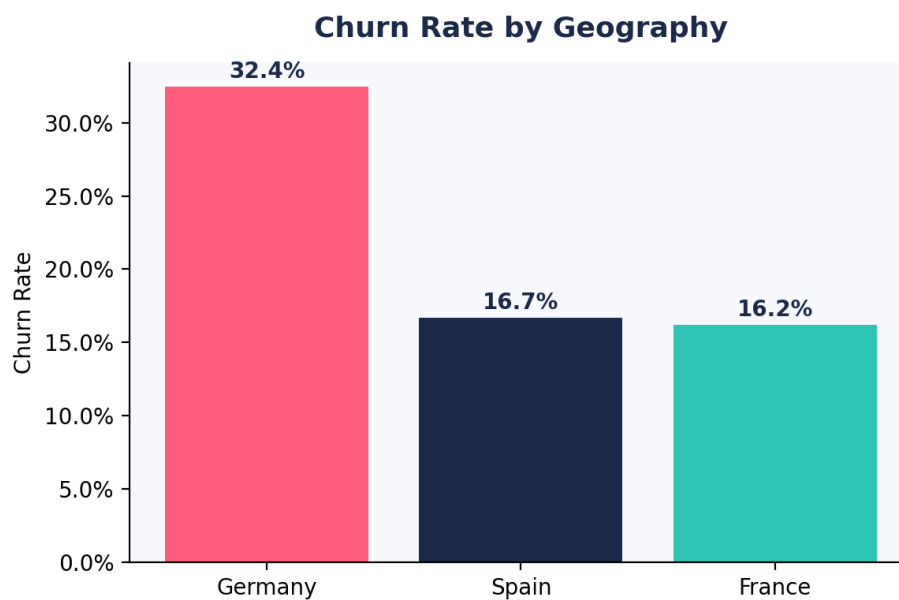


Figure 1. German customers churn at a notably higher rate than French or Spanish customers.

Churn Rate by Number of Products

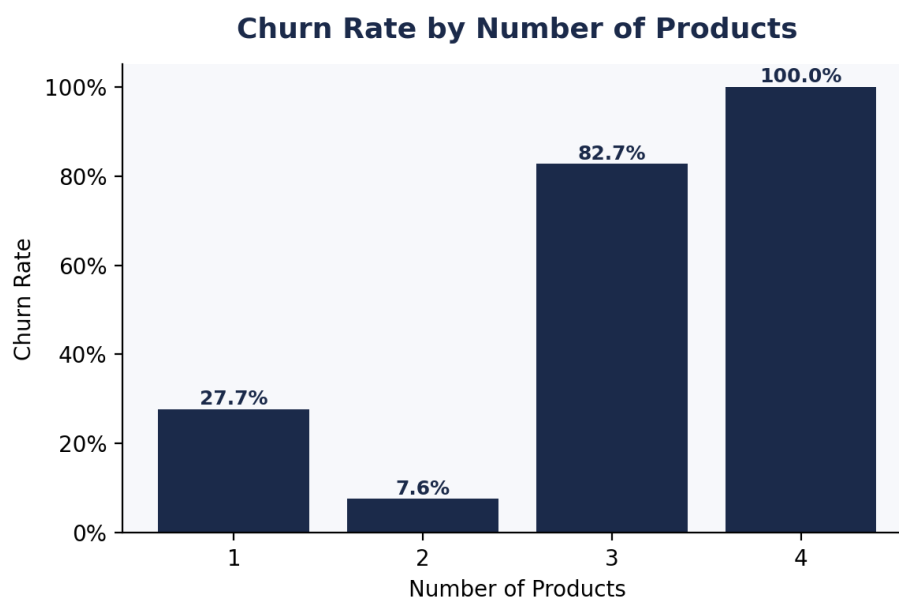


Figure 2. Customers holding 3 or 4 products churn at a dramatically higher rate, suggesting over-selling or dissatisfaction among heavy-product users.

Age Distribution by Churn Status

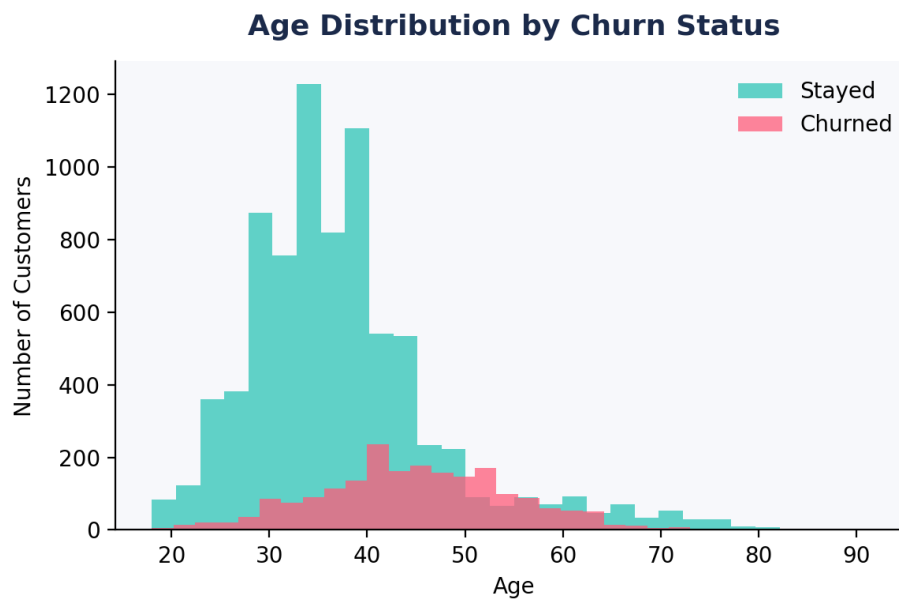


Figure 3. Churned customers skew older on average than retained customers.

Feature Correlation Matrix

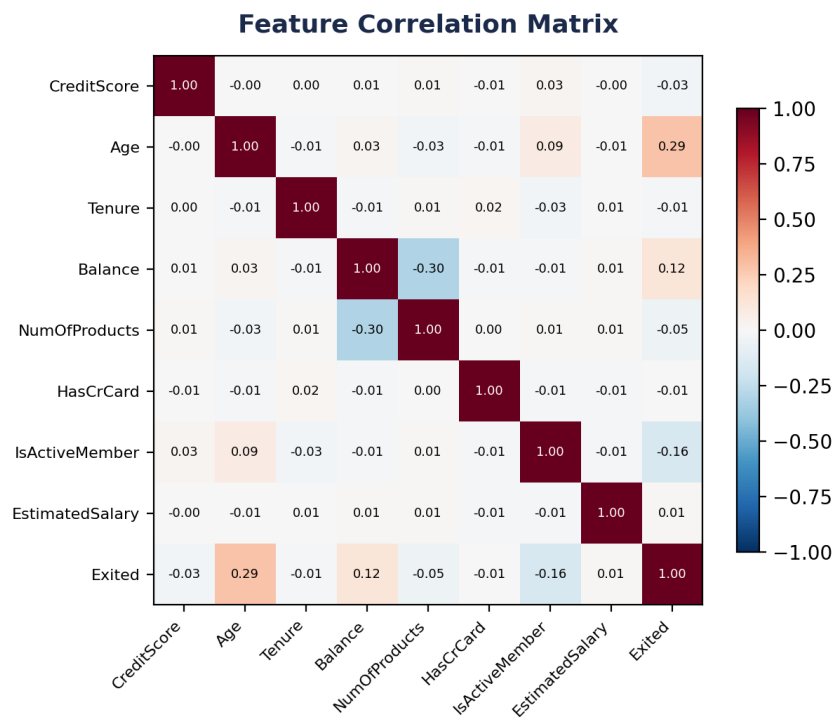


Figure 4. Age shows the strongest positive correlation with churn among numerical features; correlations overall are modest, motivating a non-linear model.

05 Methodology & Preprocessing

The raw data was transformed into a model-ready format through the following pipeline:

- **Feature separation** — numerical features (CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary) were separated from categorical features (Geography, Gender).
- **One-hot encoding** — categorical features were encoded using scikit-learn's OneHotEncoder, expanding Geography and Gender into binary indicator columns.
- **Feature scaling** — all resulting features were standardized with StandardScaler (zero mean, unit variance) so that no single feature dominates the network's gradient updates.
- **Class balancing with SMOTE** — since only ~20% of customers churned, the model would otherwise be biased toward predicting the majority class. SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic examples of the minority (churned) class until both classes are equally represented.
- **Train / validation / test split** — the balanced dataset was split into training, validation, and test sets, with the validation set used for early stopping and learning-rate scheduling.

Class Balance Before and After SMOTE

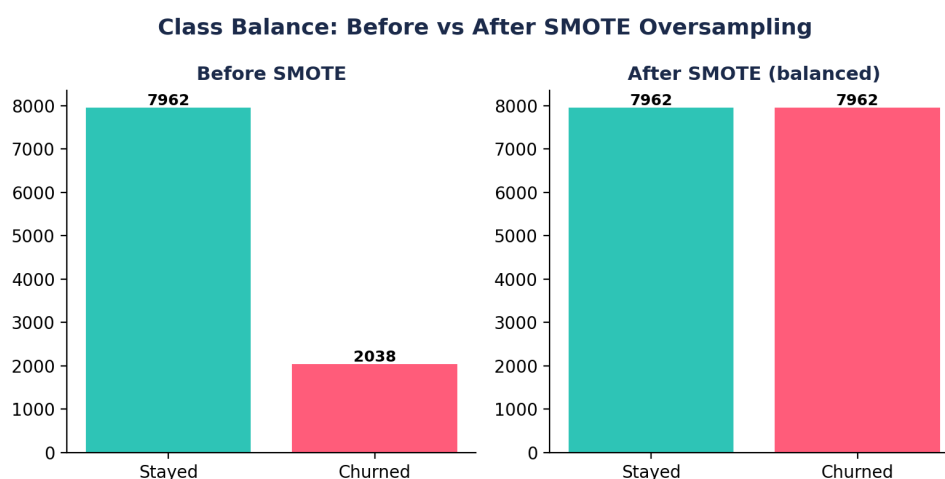
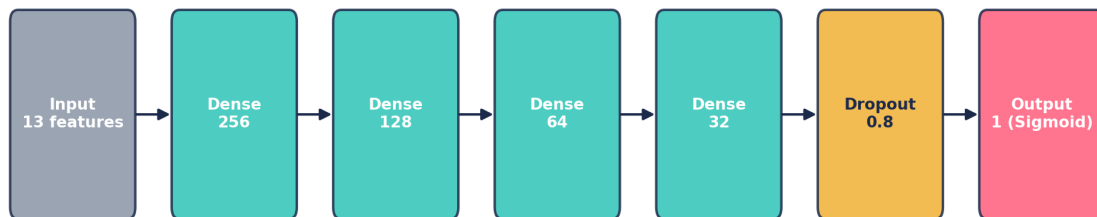


Figure 5. SMOTE oversampling equalizes the churned and retained classes before training.

06 Model Architecture

The classifier is a fully-connected feed-forward neural network implemented in PyTorch. The architecture progressively compresses the 13-dimensional input (8 numerical + 5 one-hot encoded categorical features) down to a single churn probability.

Neural Network Architecture



Layer	Output Shape	Activation
Input	13	—
Dense (Linear)	256	ReLU
Dense (Linear)	128	ReLU
Dense (Linear)	64	ReLU
Dense (Linear)	32	ReLU
Dropout (p = 0.8)	32	—
Dense (Linear, Output)	1	Sigmoid

A notably high dropout rate (0.8) is applied before the final layer as a strong regularizer, reducing the risk of overfitting on the relatively small, SMOTE-augmented training set.

07 Training Configuration

Hyperparameter	Value
Loss function	Binary Cross-Entropy (BCELoss)
Optimizer	Adam
Initial learning rate	2e-3
LR scheduler	ReduceLROnPlateau (factor 0.4, patience 20, min LR 1e-3)
Batch size	400
Max epochs	100
Early stopping patience	20 epochs (best validation loss restored)

Training used mini-batch gradient descent with early stopping: the model's weights were checkpointed whenever validation loss improved, and training halted once no improvement was observed for 20 consecutive epochs. This prevents overfitting while allowing the network to converge fully on the balanced training set.

08 Results & Evaluation

The trained model was evaluated on a held-out test set (1,592 samples, drawn from the SMOTE-balanced pool) that the model never saw during training or validation.

Test Set Metrics

Metric	Score	Interpretation
Accuracy	89.6%	Overall proportion of correct predictions
ROC AUC	89.6%	Model's ability to separate churn vs. non-churn classes
Precision	84.6%	Of customers flagged as churn risks, 84.6% actually churned
Recall	96.5%	Of customers who actually churned, 96.5% were correctly identified

Confusion Matrix

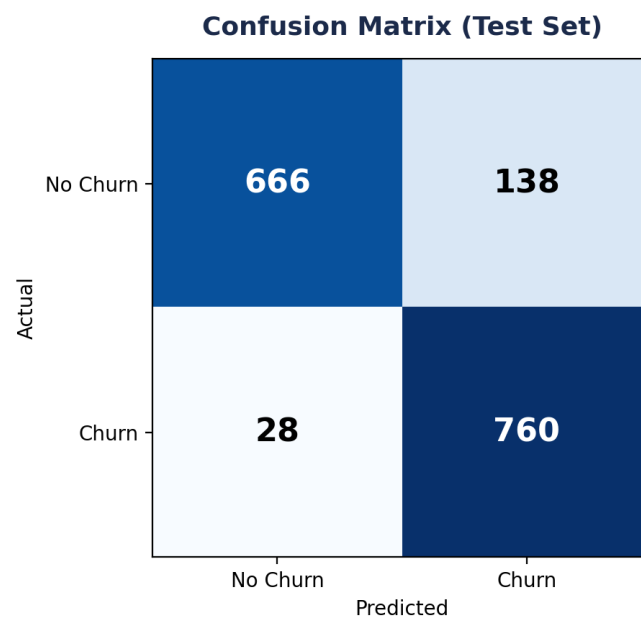


Figure 6. 760 true positives and 666 true negatives against 138 false positives and only 28 false negatives.

The high recall (96.5%) is particularly valuable in a churn-prevention context: missing an actual churner (a false negative) is far more costly to the business than mistakenly flagging a loyal customer for outreach (a false positive). The low false-negative count (28) shows the model is well-tuned for this priority.

09 Application Deployment

The trained model, encoder, and scaler were serialized and wrapped in an interactive **Streamlit** web application (`app.py`), giving non-technical stakeholders direct access to the model without needing to touch code or notebooks.

Application Features

Tab	Functionality
Prediction	Single-customer churn probability with a risk gauge and comparison to the average customer profile
Batch	CSV upload for scoring many customers at once, with a downloadable results file
Data Insights	Live charts: churn rate by geography, correlation heatmap, and feature distributions
About	Model architecture, preprocessing summary, and stated limitations

A configurable decision threshold lets the user tune the sensitivity of the churn/stay classification depending on business priorities (e.g., a lower threshold catches more at-risk customers at the cost of more false alarms).

Technology Stack

- **PyTorch** — model definition, training, and inference
- **scikit-learn** — preprocessing (encoding, scaling) and evaluation metrics
- **imbalanced-learn (SMOTE)** — class balancing
- **Streamlit** — web application framework
- **Plotly** — interactive charts within the app

10 Limitations

- **Probability calibration** — because training data was rebalanced with SMOTE, the model's output probabilities do not reflect the true real-world churn base rate (~20%) and should be treated as relative risk scores rather than calibrated probabilities.
- **High dropout noise** — the 0.8 dropout rate, while effective for regularization, can introduce some variance into individual predictions; batch or ensembled predictions are more stable than single-sample outputs.
- **Static feature set** — the model relies solely on the ten input features available in the dataset and does not incorporate behavioral or transactional time-series data (e.g., recent transaction frequency).
- **Not a substitute for full retention analysis** — this project is intended as a demonstration and portfolio piece; production use would require calibration, fairness audits, and integration with real-time data pipelines.

11 Conclusion & Future Work

This project demonstrates a complete, reproducible machine learning workflow for a real-world business problem: from raw tabular data, through preprocessing and class balancing, to a trained neural network and a usable deployment. The resulting model achieves strong recall (96.5%) on identifying at-risk customers, making it well-suited to a retention use case where missing a churner is costlier than a false alarm.

Potential Improvements

- Calibrate output probabilities (e.g., with Platt scaling) to reflect the true churn base rate.
- Compare the neural network against gradient-boosted tree baselines (XGBoost, LightGBM) for accuracy and interpretability trade-offs.
- Add explainability tooling (e.g., SHAP values) so retention teams understand *why* a customer is flagged.
- Incorporate additional behavioral features, such as transaction recency and support-ticket history.
- Containerize and deploy the Streamlit app to a cloud platform (e.g., Streamlit Community Cloud, Render, or AWS) for live use.

End of Report — Bank Customer Churn Prediction Project